Math 340 • Fall 2026 • Victor Matveev

5.5 Gauss-Legendre Quadrature

Newton-Cotes methods such as Simpson's rule or Boole's rule have impressive accuracy, and are easy to code. Moreover, using Richardson extrapolation, we can achieve arbitrarily high precision using linear combinations of simpler rules.

And yet, we know that these methods can't be optimal, since they are equivalent to integrating the integrand's interpolating polynomial with equidistant interpolation nodes. We learned that the accuracy of polynomial interpolation increases if we use non-equidistant nodes. For instance, Chebyshev nodes guarantee near-optimal mini-max interpolation error.

In fact, we find (see next page) that the optimal *n*-point quadrature rules involve irrational fractions of integration interval. Such optimal integration approximations are called Gauss Quadrature or Gauss-Legendre Quadrature:

$$\int_{-h}^{h} f(x)dx \;\approx\; \mathbf{G}_2[\mathbf{f}] = h\left[f\left(-\frac{h}{\sqrt{3}}\right)+f\left(+\frac{h}{\sqrt{3}}\right)\right] :\ \text{degree of precision} = 3.$$

$$\int_{-h}^{h} f(x)dx \;\approx\; \mathbf{G}_3[\mathbf{f}] = \frac{h}{9}\left[5f\left(-h\sqrt{\frac{3}{5}}\right)+8f(0)+5f\left(+h\sqrt{\frac{3}{5}}\right)\right] :\ \text{degree of precision} = 5.$$

$$\int_{-h}^{h} f(x)dx \;\approx\; \mathbf{G}_4[\mathbf{f}] = h\frac{18-\sqrt{30}}{36}\left[f\left(-h\sqrt{\frac{3+2\sqrt{6/5}}{7}}\right)+f\left(+h\sqrt{\frac{3+2\sqrt{6/5}}{7}}\right)\right]$$
$$+h\frac{18+\sqrt{30}}{36}\left[f\left(-h\sqrt{\frac{3-2\sqrt{6/5}}{7}}\right)+f\left(+h\sqrt{\frac{3-2\sqrt{6/5}}{7}}\right)\right] :\ \text{degree of precision} = 7.$$

$$\int_{-h}^{h} f(x)dx \;\approx\; \mathbf{G}_5[\mathbf{f}] = h\frac{322-13\sqrt{70}}{900}\left[f\left(-\frac{h}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}\right)+f\left(+\frac{h}{3}\sqrt{5+2\sqrt{\frac{10}{7}}}\right)\right]$$
$$+h\frac{322+13\sqrt{70}}{900}\left[f\left(-\frac{h}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}\right)+f\left(+\frac{h}{3}\sqrt{5-2\sqrt{\frac{10}{7}}}\right)\right]+\frac{128h}{225}f(0) :\ \text{degree of precision} = 9.$$

The degree of precision of Gauss quadrature with $\boldsymbol{n}$ points equals $\boldsymbol{2n-1}$, the highest possible for such rules.

If the interval of integration $[a,b]$ is not centered at zero, we have to shift all nodes by the mid-point, $x_{mid} = \frac{a+b}{2}$.

Compare this with the elementary Simpson's or Boole's methods with $\boldsymbol{n}$ function evaluations, for which the degree of precision is just $\boldsymbol{n}$ (i.e. 3 and 5, respectively).

The error expression makes it clear just how quickly the accuracy of this method ramps up with each new added node:

$$\left|\text{Error}\left(G_m[f]\right)\right| = \left|\frac{(b-a)^{2m+1}(m!)^4}{(2m+1)\left[(2m)!\right]^3}f^{(2m)}(c)\right| \;\Rightarrow\; \begin{cases} \left|\text{Error}\left[G_2\right]\right| = \left|\dfrac{f^{(4)}(c)}{135}h^5\right| \\ \left|\text{Error}\left[G_3\right]\right| = \left|\dfrac{f^{(6)}(c)}{15{,}750}h^7\right| \\ \left|\text{Error}\left[G_4\right]\right| = \left|\dfrac{f^{(8)}(c)}{3{,}472{,}875}h^9\right| \\ \left|\text{Error}\left[G_5\right]\right| = \left|\dfrac{f^{(10)}(c)}{1{,}237{,}732{,}650}h^{11}\right| \end{cases} \quad \text{where } c\in(-h,h).$$

As with Newton-Cotes formulas, the caveat is that the integrand has to be continuously differentiable $2m$ times in order to achieve such high accuracy: $f(x) \subset C^{2m}_{[-h,h]}$.

We can form composite Gauss quadrature by sub-dividing the integration interval, but note that the advantage of Gauss quadrature is reduced in this case, as compared to Newton-Cotes methods supplemented with Richardson extrapolation.

Derivation of 2-point Gauss Quadrature

In class, we derived Gauss quadrature by optimizing the weights and nodes so that maximal degree of precision is achieved. Let's reproduce this derivation here.

Start by considering a general 2-point rule. Without loss of generality, we will pick the integration interval to be $[-h, h]$ (in class we considered $h = 1$, for the sake of simplicity). In general, symmetric weights and nodes guarantee exact zero result for any odd polynomial, so an optimal rule has to be symmetric:

$$\int_{-h}^{h} f(x)dx \approx G_2 = w\left[f(-x_*) + f(x_*)\right]$$

Since there are two unknowns, x_* and w, we can enforce zero error for two even monomials, $f(x) = 1$ and $f(x) = x^2$:

$f(x)$	Approximation $G_2[f] \equiv w\left[f(-x_*) + f(x_*)\right]$	Exact integral $I[f] \equiv \int_{-h}^{h} f(x)dx$	Solution for Error = 0 or (for $n=4$) error itself, $G_2\left[x^4\right] - I\left[x^4\right]$
1	$2w$	$\int_{-h}^{h} dx = 2h$	$\boxed{w = h}$
x	$w\left[-x_* + x_*\right] = 0$	$\int_{-h}^{h} x\,dx = 0$	
x^2	$w\left[(-x_*)^2 + (x_*)^2\right] = 2w\cdot(x_*)^2$	$\int_{-h}^{h} x^2dx = \frac{2h^3}{3}$	$w\cdot(x_*)^2 = h\cdot(x_*)^2 = \frac{h^3}{3} \Rightarrow \boxed{x_* = \pm\frac{h}{\sqrt{3}}}$
x^3	$w\left[(-x_*)^3 + (x_*)^3\right] = 0$	$\int_{-h}^{h} x^3dx = 0$	
x^4	$w\left[(-x_*)^4 + (x_*)^4\right] = 2w\cdot(x_*)^4$	$\int_{-h}^{h} x^4dx = \frac{2h^5}{5}$	$2\left[w\cdot(x_*)^4 - \frac{2h^3}{5}\right] = 2\left[\frac{h^5}{9} - \frac{h^5}{5}\right] = \frac{8h^5}{45}$

Thus, the error of this rule is:

$$Error\left[f(x)\right] = Error\left[f(0)\cdot 1 + f'(0)\cdot x + \ldots + \frac{f^{(4)}(c_x)}{4!}\cdot x^4\right] = \frac{f^{(4)}(C)}{4!}\cdot \underbrace{Error\left[x^4\right]}_{8h^5/45} = \boxed{\frac{f^{(4)}(C)}{135}h^5} \text{ where } C \in (-h, h)$$

EXAMPLE: Use 2-point Gauss quadrature to approximate $\int_1^2 \frac{dx}{x} = \ln(2) \approx \mathbf{0.6931472}$. .

Solution: $$G_2 = h\left[f\left(x_{mid} - \frac{h}{\sqrt{3}}\right) + f\left(x_{mid} + \frac{h}{\sqrt{3}}\right)\right] = \frac{1}{2}\left[\frac{1}{\frac{3}{2} - \frac{1}{2\sqrt{3}}} + \frac{1}{\frac{3}{2} - \frac{1}{2\sqrt{3}}}\right] = \frac{1}{3 - \frac{1}{\sqrt{3}}} + \frac{1}{3 - \frac{1}{\sqrt{3}}} = \frac{6}{9 - \frac{1}{3}} = \frac{18}{26} = \frac{\mathbf{9}}{\mathbf{13}} \approx \mathbf{0.69230}.$$

Actual error: $G_2 - \ln(2) \approx -\mathbf{0.00084}$.

-- significantly better than Simpson's rule that uses **3** function values: $S_2 - \ln(2) = \frac{25}{36} - \ln(2) = \mathbf{0.0013}$.

Theoretical error bound: $$\left|\frac{f^{(4)}(C)}{135}h^5\right| = \left|\frac{24/C^5}{135}\left(\frac{1}{2}\right)^5\right| \le \left|\frac{24}{135\cdot 32}\right| = \left|\frac{3}{135\cdot 4}\right| \approx \mathbf{0.00555}\ldots$$

Legendre polynomials

Although we obtained Gauss quadratures in class using straightforward maximization of degree of precision, the nodes we ended up with turn out to be connected with zeros of Legendre polynomials, defined as:

$$L_n(x) = \frac{1}{n!\,2^n} \cdot \frac{d^n}{dx^n}(x^2-1)^n$$

where the constant pre-factor is not that important (it ensures that $L_n(1) = 1$).

Using this formula you can check that the first several polynomials are given by

$$L_0(x) = 1$$

$$L_1(x) = x$$

$$L_2(x) = \frac{1}{2}(3x^2-1)$$

$$L_3(x) = \frac{1}{2}x(5x^2-3)$$

Note: the zeros of L_1, L_2, L_3 agree respectively with the nodes of integration rules $G_1 = M_1$ (midpoint rule), G_2 and G_3.

Orthogonality of Legendre Polynomials

An important property of this polynomial set is that they are mutually orthogonal on the interval [-1, 1], in the sense of the "inner product" defined as the integral of a product of two functions (this allows to treat functional spaces as vector spaces, in which this inner product generalizes the dot product of two vectors):

$$\int_{-1}^{1} L_k(x)L_m(x)dx = 0 \text{ if } k \neq m.$$

For instance, it is easy to check that

$$\int_{-1}^{1} L_1(x)L_3(x)dx = \frac{1}{2}\int_{-1}^{1} x^2(5x^2-3)dx = \frac{1}{2}\left[\frac{5x^5}{5} - \frac{3x^3}{3}\right]_{-1}^{1} = 0$$

In general, the proof of orthogonality for general *k* and *m* is easy to obtain using integration by parts, starting with the definition of $L_n(x)$ given above, and following from the fact that $(x^2-1) = 0$ at the endpoints $x = \pm 1$.

Further, it follows that $L_n(x)$ is orthogonal to any polynomial of degree $\boldsymbol{l < n}$, since any polynomial can be expressed as a linear combination of any set of polynomials of same degree and lower, that is

$$P_l(x) = \sum_{k=0}^{l<n} c_k L_k(x)$$

Orthogonality of $P_{l<n}(x)$ and $L_n(x)$ then follows from the linearity of integration:

$$\int_{-1}^{1} L_n(x)P_l(x)dx = \int_{-1}^{1} L_n(x)\left[\sum_{k=0}^{l<n} c_k L_k(x)\right]dx = \sum_{k=0}^{l<n} c_k\left[\int_{-1}^{1} L_n(x)L_k(x)\right]dx = 0$$

Connection between optimal quadrature and Legendre polynomials

We are now ready for the proof that Gauss quadrature nodes are optimal, in the sense that it is exact for degree $2n-1$ polynomials (we can't expect more, as that would require more constraints that the number of weights and nodes).

Note that any polynomial of degree $\boldsymbol{m = 2n-1}$ (for which we want the method to be exact) can be written in the form:

$$P_m(x) \equiv P_{2n-1}(x) = L_n(x)p_{n-1}(x) + q_{n-1}(x)$$

where $q_{n-1}(x)$ is the remainder obtained by dividing polynomial $P_{2n-1}(x)$ by the Legendre polynomial $L_n(x)$.

We will now show that the *n*-point quadrature that uses zeros of Legendre polynomial, $L_n(x_k) = 0$, is exact when applied to $P_{2n-1}(x)$.

Let's first show that the rule is exact for the first term of the above decomposition of polynomial $P_{2n-1}(x)$:

$$\text{Error}\left[L_n(x)p_{n-1}(x)\right] = G_n\left[L_n(x)p_{n-1}(x)\right] - \int_{-1}^{1} L_n(x)p_{n-1}(x)dx =$$

$$= \sum_{k=1}^{n} w_k\left[L_n(x_k)p_{n-1}(x_k)\right] - \int_{-1}^{1} L_n(x)p_{n-1}(x)dx = 0$$

The zero error follows because:

- The first term above is zero since x_k are by definition zeros of $L_n(x)$
- The second part of the error is zero by orthogonality of polynomials $L_n(x)$ and $p_{m<n}(x)$, proven above.

We only have to show that the error is also zero for the "remainder", $q_{n-1}(x)$. Note however that requiring zero error for an arbitrary polynomial of degree $n-1$ in fact gives us n equations for the n unknown quadrature weights w_k!

In fact, for any n-point method, weights w_k are found by integrating interpolating polynomial over n nodes x_k, which guarantees zero error for any polynomial of degree $n-1$.

This concludes the proof ☺.

The only thing we didn't prove or derive is the general expression for the **weights** for the n-point Gauss quadrature, which can be shown to equal:

$$w_k = \frac{2}{\left(1-x_k^2\right)\left[L_n'(x_k)\right]^2}, \quad k = 1..n.$$

For instance, for $n = 2$ the Legendre polynomial is $L_2(x) = \frac{1}{2}(3x^2 - 1)$, with zeros $x_{1,2} = \pm\frac{1}{\sqrt{3}}$, and $L_2'(x) = 3x$, therefore:

$$w_k = \frac{2}{\left(1-\frac{1}{3}\right)\left[\pm\frac{3}{\sqrt{3}}\right]^2} = \frac{2}{2} = 1.$$

Why not Chebyshev Nodes?

You may wonder why the optimal integration nodes are not Chebyshev nodes: after all, the quadrature rules we started with can be obtained by integrating the corresponding polynomial interpolant over the quadrature points, and we learned that Chebyshev interpolation is near-optimal. Recall however that the best interpolation is the one with the smallest maximal absolute error (the "mini-max" error, or the smallest value of the sup-norm, also known as infinity-norm), while the integration error is the *integral* of the interpolation error: strictly speaking, integral of the error is not an error norm in the mathematical sense, since it can be either positive or negative. Because of the different goals, we obtain different optimal nodes.

It is interesting however that Chebyshev nodes **are** in fact the optimal nodes for numerical approximation of integrals with singular integrands of the following form:

$$\int_{-1}^{1} \frac{f(x)dx}{\sqrt{1-x^2}}$$

This is because Chebyshev polynomials **are** in fact **orthogonal** over the interval [-1, 1] with respect to integrals with integration measure weighted by $\frac{1}{\sqrt{1-x^2}}$.